Hematology & Hemodynamics

The Triage Mechanist's Field Manual

Chief Mechanist

Operating the Human Engine Under Pressure.

NotebookLM

STABLE
EXPECTED
(Monitor)
CAUTION
CHRONIC
(Be Concerned)
DANGER
ACUTE
DEADLY
(Action STAT)
MECHANIST
INTERVENTION
(Nursing Action)
INTERNAL
MECHANICS
(Pathophysiology)
Memorize the lights. They dictate survival.
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THE HYDRAULIC PUMP (Cardiac Status)

The engine drives the system. We must monitor the stress on the metal (Troponin) and the internal pressure of the chambers (BNP).

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BNP: THE STRETCH GAUGE

Mechanism: B-type Natriuretic Peptide released when ventricles STRETCH due to fluid overload.

Normal: < 100

Elevated: > 100 = CHF (Chronic Heart Failure)

Mechanist Note: Indicates a chronic condition. Monitor, but do not panic. Not an acute emergency.

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TROPONIN: STRUCTURAL FAILURE

Mechanism: Proteins released only when heart muscle dies (Ischemia/Infarction).

Elevated Troponin is NEVER normal.

Priority: INDICATES ACTIVE MYOCARDIAL INFARCTION.

Action: Treat as ‘Level D’ Priority. The engine is melting down.

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STEAM REGULATION (Acid-Base Balance)

pH 7.35 - 7.45: Optimal Pressure (Green)

pH in the 6s: CRITICAL FAILURE (Deadly D). The boiler will explode.

Action: If pH in 6s: Assess Vitals, Call MD STAT. Stay with the patient.

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DIAGNOSTIC PROTOCOL: R.O.M.E.

Check pH.
(Acid < 7.35 | Base > 7.45)

Check Lung (CO_2).
Is it the cause?

Respiratory Opposite
(pH up, CO_2 down)

Check Kidney (HCO_3).
Is it the cause?

Metabolic Equal
(pH up, HCO_3 up)

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Mac-Kussmaul's Respirations

Mechanism: The body is trying to blow off acid (CO_2).

Rule: Kussmaul Respirations = MAC (Metabolic ACidosis).

You do **NOT** see Kussmaul with Respiratory Acidosis. Only **Metabolic**.

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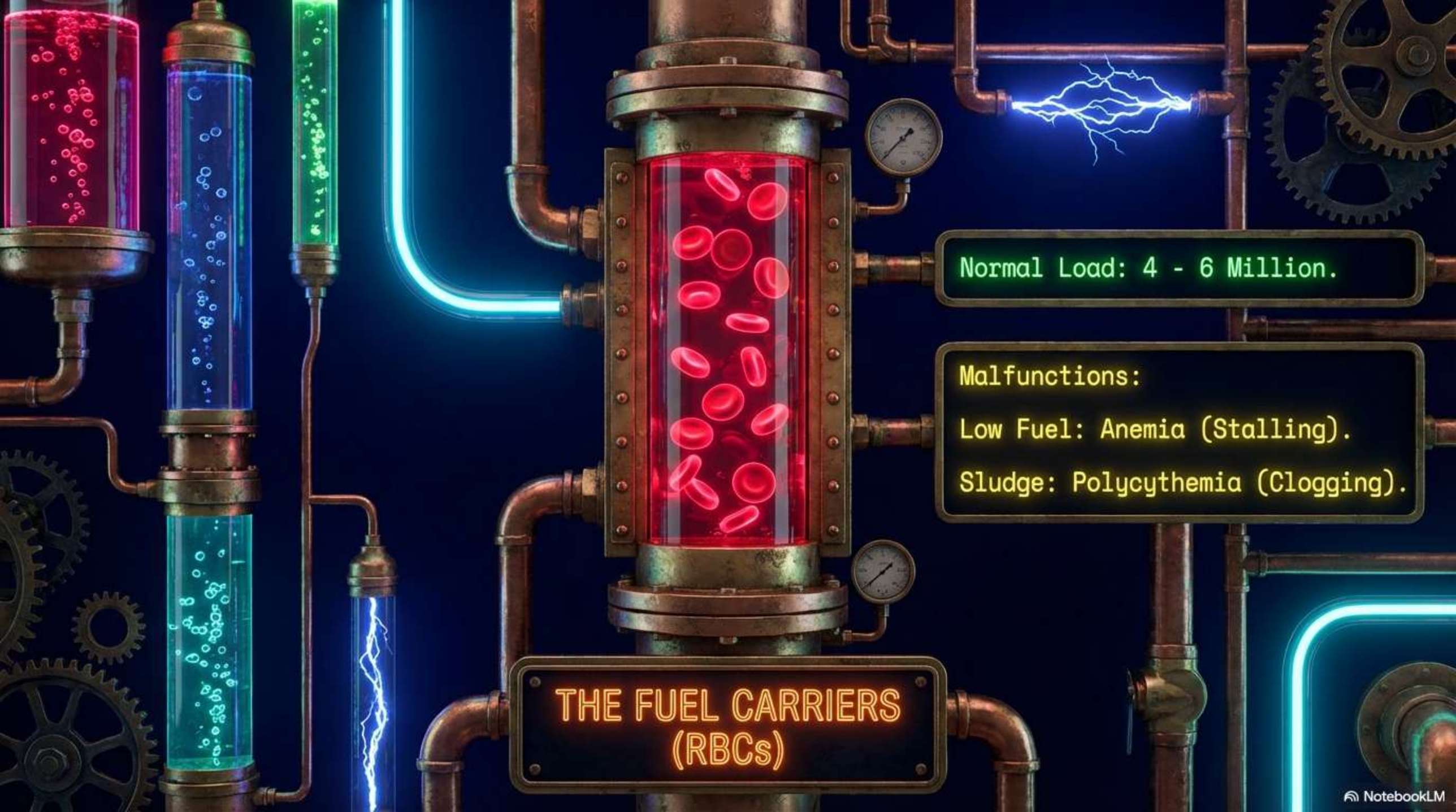
THE FUEL CARRIERS
(RBCs)
Normal Load: 4 - 6 Million.
Malfunctions:
Low Fuel: Anemia (Stalling).
Sludge: Polycythemia (Clogging).

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ANEMIA DIAGNOSTICS

Symptoms: Fatigue, Pallor, Dyspnea on Exertion.

Mechanist Intervention: PACE ACTIVITIES. Do not over-rev the engine. Allow for rest periods.

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PERNICIOUS ANEMIA: THE BROKEN GEAR

Mechanism: Stomach lacks Intrinsic Factor. Cannot absorb Vitamin B12.

Symptoms: Beefy Red Tongue, Paresthesia (Numbness/Tingling).

Treatment: Oral Iron will NOT work. Requires lifelong IM B12 Injections.

Diagnostic: Schilling Test.

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POLYCYTHEMIA VERA: VISCOSITY WARNING

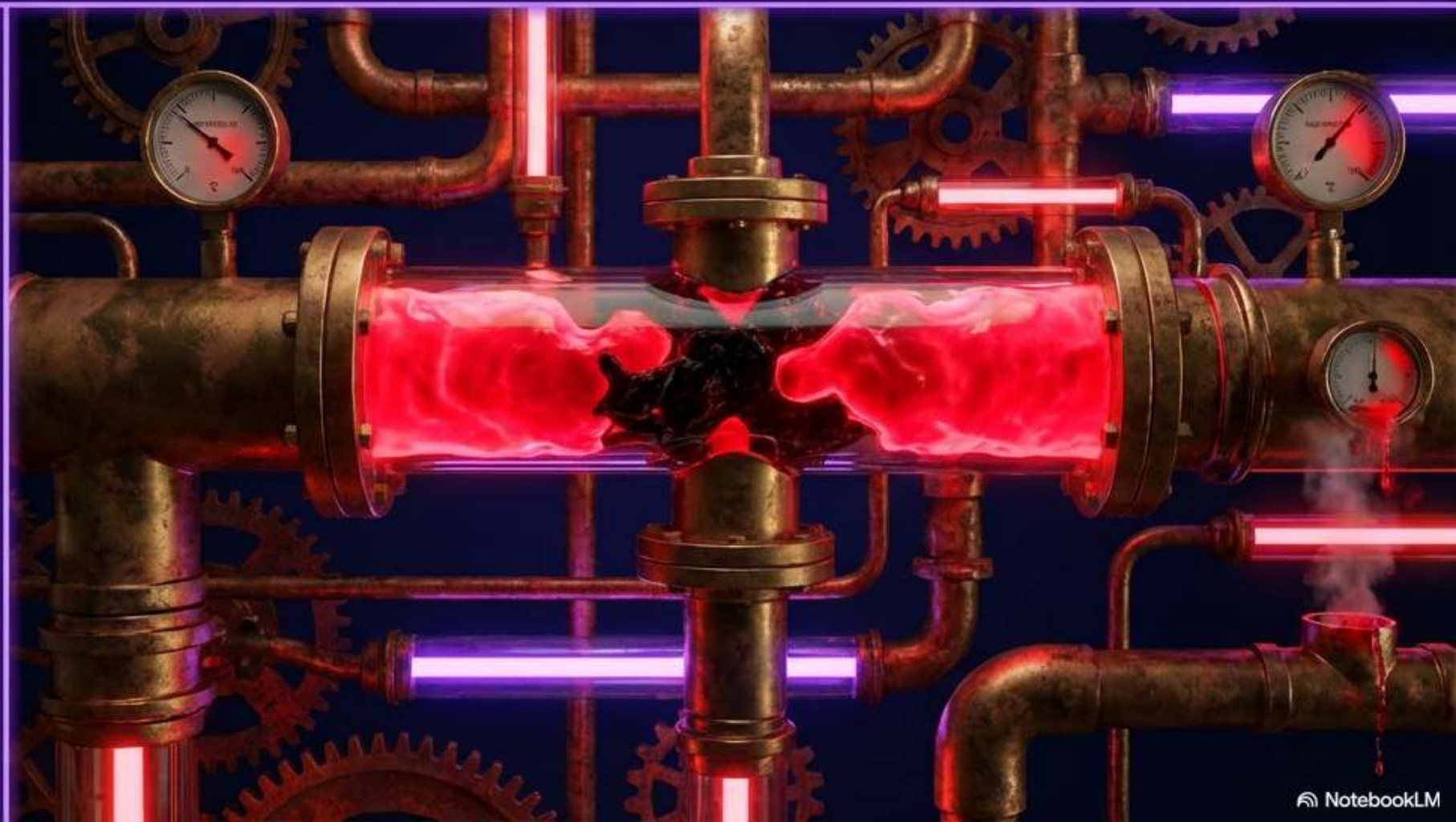
Danger: Thrombus (Clot) formation due to excessive RBCs.

Protocol:

HYDRATION:
> 3 Liters/day to thin the blood.

DIET:
Low Iron (Avoid spinach, liver).

ACTION:
Phlebotomy (Drain the excess).



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SICKLE CELL CRISIS: THE JAMMED COG

Priority Protocol: H.O.P.

1. Hydration (IV Fluids to unclog).
2. Oxygen (To fuel tissues).
3. Pain Relief (To comfort the operator).

Mechanism:

Heat vasodilates (unclogs).
Heat vasocnstricts (bad).
Cold vasoconstricts (bad).

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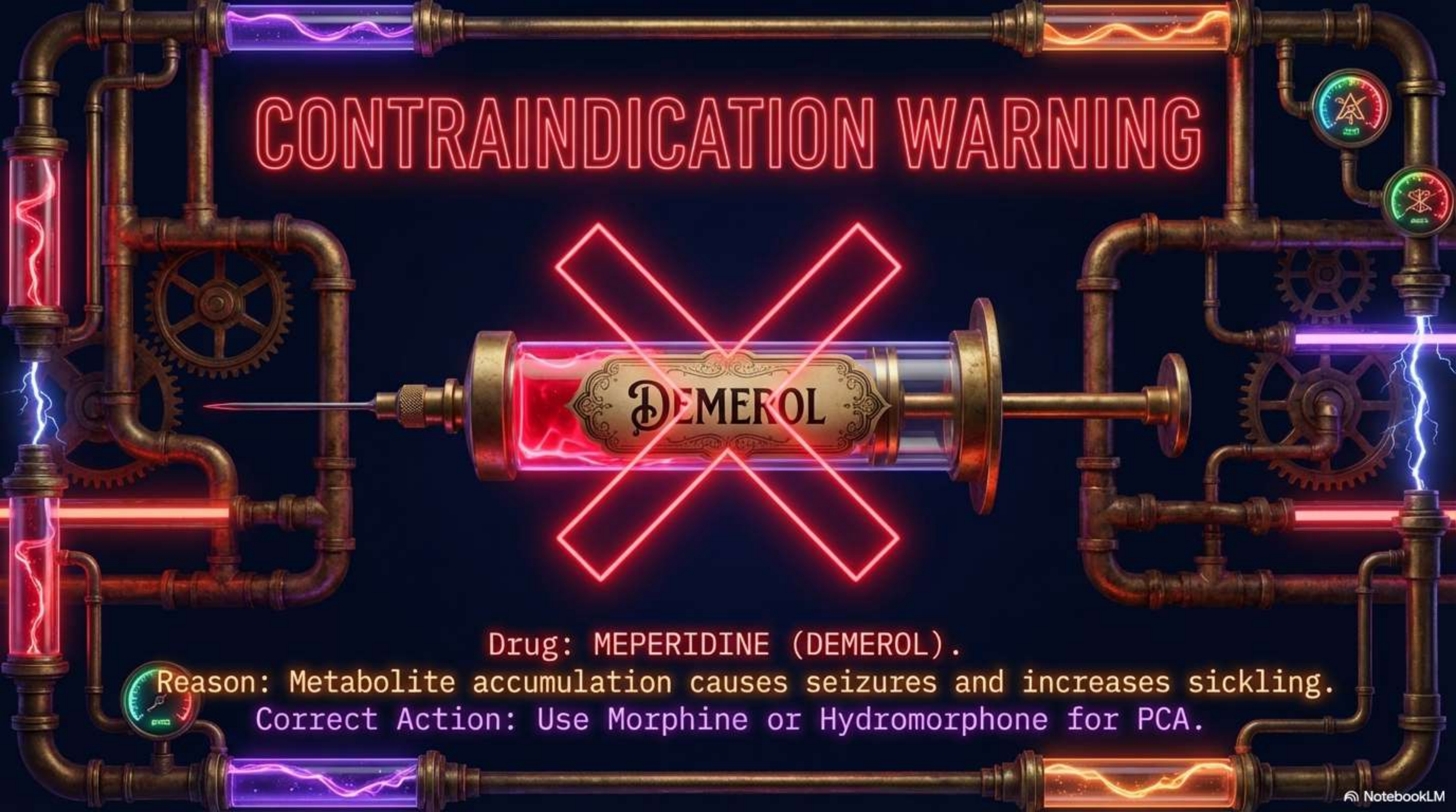
CONTRAINDICATION WARNING



Drug: MEPERIDINE (DEMEROL).

Reason: Metabolite accumulation causes seizures and increases sickling.

Correct Action: Use Morphine or Hydromorphone for PCA.

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THE CLOTTING GEARS: HEPARIN

- Speed: Works Immediately (IV/SubQ).
- Duration: Short term (Max 21 days).
- Lab Gauge: PTT.
- Mnemonic: Heparin + PTT = 10 fingers.
- Antidote: Protamine Sulfate.
- Pregnancy: Safe (Class C).

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THE CLOTTING GEARS: WARFARIN

- Speed: Takes days to work.
- Route: PO Only.
- Lab Gauge: PT/INR.
- Mnemonic: Coumadin + PT = 8 fingers.
- Antidote: Vitamin K.
- Pregnancy: DANGEROUS (Class X). Never give to pregnant women.

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THE DEFENSE GRID (WBCs)

Safe Range: Total WBC 5,000 - 11,000.

Grid Failure:

- WBC < 5,000
- ANC (Neutrophils) < 500
- CD4 < 200 (AIDS Threshold)

Status: IMMUNOSUPPRESSION (Level C Priority).

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NEUTROPENIC PROTOCOL

- Strict handwashing. Shower BID with antimicrobial soap.
- Dedicated Equipment (Stethoscope & BP Cuff) stays in room.
- Isolation: Private room. Limit staff/visitors.

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ENVIRONMENTAL HAZARDS

The Rule: If it is alive or stagnant, it carries bacteria.

- NO Fresh Flowers.
- NO Potted Plants.
- NO Raw Fruits/Veggies (Low bacteria diet).
- NO Undercooked Meat.
- NO Water pitcher standing > 15 Minutes.

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THE DEADLY D's (Highest Priority)

1. pH in the 6s.
2. Potassium > 6.
3. CO2 in the 60s.
4. O2 in the 60s.
5. Platelets < 40,000.

ACTION: STAY WITH THE PATIENT. CALL FOR HELP.

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PROTOCOLS COMPLETE.

The Engine is Stabilized.

Shift Change.

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